

Samir Bhusal

samir.masterprep@gmail.com | bhusalsamir.com.np | github.com/samirbhusal

Education

Dakota State University | Madison, SD

August 2025 – Present

Master of Science, Information Systems

Purbanchal University | Kathmandu, Nepal

October 2017 – October 2022

Bachelor of Science, Computer Applications

Project: Created ecommerce platform with product categories, search, add to cart, payment integration (Khalti) using python Django Framework and SQLite.

Experience

Graduate Research Assistant | Dakota State University, Madison, SD

August 2025 – Present

- Designed an interactive [Figma prototype](#) (“Helpdesk RAG Assistant”) showcasing an AI-powered incident resolution system and performance dashboard and presented at the Siouxland IT Symposium.
- Tools/Skills:** Figma, Wireframing, Prototyping, UX/UI Design, Interaction Design, Smart Animate, AI Product Design, Dashboard Visualization, User Flow
- Supported faculty research in XAI Healthcare by building and maintaining a literature matrix.
- Assisted in evaluating Python-based analytics projects and assignments by making rubric and grading

Software Engineer – Quality Assurance | Deerhold Ltd., Kathmandu, Nepal

November 2022 – August 2025

- Maintained automated test coverage of more than 90 percent and pass rate over 95 percent for [Higi platforms](#) using BDD Cucumber framework with Selenium, Appium and Java for mobile and web platforms with reduced release cycle by 30%.
- Developed and maintained test automation for the mobile app (Android, iOS) [\(InStride Health\)](#) with 80 percent automation coverage using BDD Cucumber framework with Selenium, Appium and Java.
- Managed QA processes and rebuild of legacy system, achieving production readiness in 3 months for [Wellness Portal](#).
- Managed test planning, execution, reporting, and product release documentation and performed manual and automation test execution.
- Collaborated in agile ceremonies ensuring alignment with cross-functional teams for continuous improvement.

Projects

Python-BDD-Automation Framework Python, Behave, Selenium, Page Object Model	
Developed a scalable Python Behave + Selenium POM framework with multi-browser & environment-driven execution.	
Full Stack RAG Chatbot React, Vite, Python, FastAPI, LangChain, Pinecone, Groq (Llama-3.3), HuggingFace	
Developed a RAG chatbot for intelligent incident query resolution using internal documents.	
Personal Portfolio Website HMTL, CSS, JavaScript, React.js, Vite, Tailwind, Vercel	
Built and deployed a responsive React (Vite) portfolio with reusable components, and Vercel CI/CD + custom domain setup.	
Memory Card Game React.js, Vite, Axios, React Hooks, TypeScript, GitHub Pages	
Developed a responsive memory game using React and TypeScript, leveraging Axios interceptors to manage asynchronous data fetching from the PokéAPI.	
CV App React.js, TypeScript, Vite, GitHub Pages	
Developed a dynamic single-page application using React and TypeScript to create professional resumes through a modular, section-based interface.	

Technical Skills

Languages: Python, JavaScript, SQL, Gherkin

Frameworks: Selenium, Appium, Karate, Cucumber (BDD), JUnit, TestNG, Behave, React, Vite

Cloud & Tools: Git, GitHub, JIRA, Confluence, BrowserStack, Android, iOS, Excel, Power BI, Postman – API testing

Methodologies: Manual Testing, Regression, Smoke, UAT, SDLC, STLC, Agile/Scrum, cross-browser and responsive testing,

Documentation: Test Plans, Test Cases, Bug Reports, Release Notes,

Achievements / Awards

- Winnter of Hult Prize Purbanchal University 2020